Chapter – 1 Basics of SAS

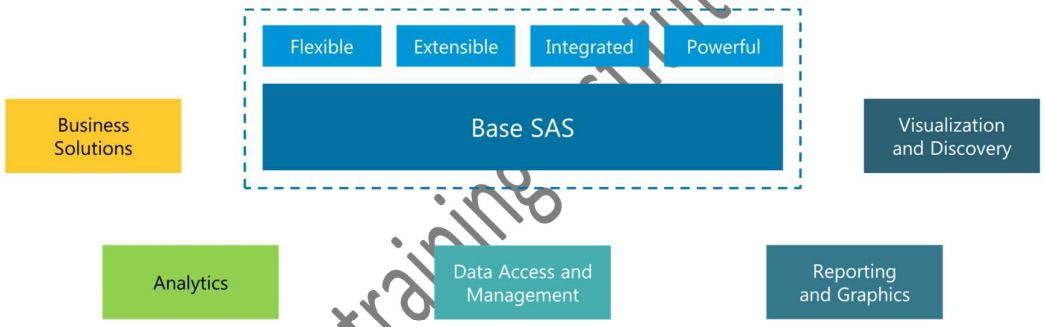
What is SAS?

Answer: SAS (**Statistical Analytics System**)

- SAS is a software suite for advanced analytics, multivariate analyses, business intelligence, data management and predictive analytics
- It is developed by SAS Institute.
- SAS provides a graphical point-and-click user interface for non-technical users and more advanced options through the SAS language.

What are the features of SAS?

Answer: The following are the features of SAS:



Features of SAS

1. **Business Solutions:** SAS provides business analysis that can be used as business products for various companies to use.
2. **Analytics:** SAS is the market leader in the analytics of various business products and services.
3. **Data Access & Management:** SAS can also be use as a DBMS software.
4. **Reporting & Graphics:** Hello SAS helps to visualize the analysis in the form of summary, lists and graphic reports.
5. **Visualization:** We can visualize the reports in the form of graphs ranging from simple scatter plots and bar charts to complex multi-page classification panels.

Mention few capabilities of SAS Framework.

Answer: The following are the four capabilities in SAS Framework:

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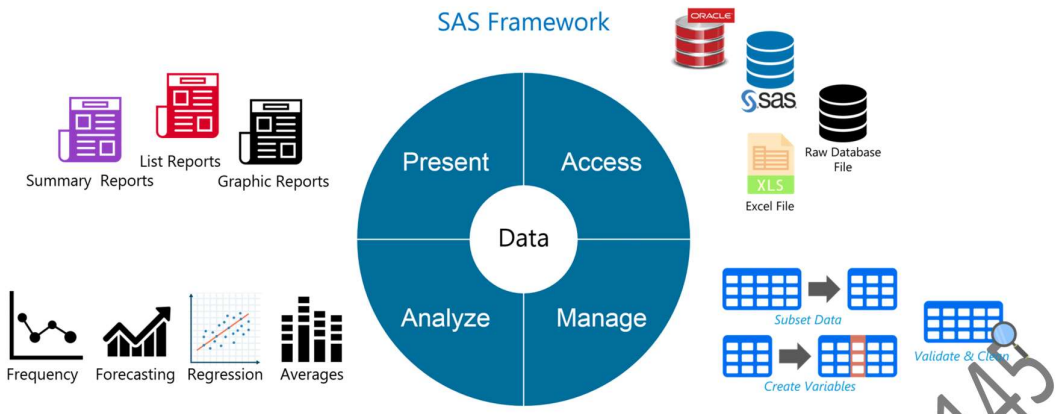

SAS Framework

1. **Access:** As we can learn from the figure, SAS allows us to access data from multiple sources like an Excel file, raw database, Oracle database and SAS Datasets.
2. **Manage:** We can then manage this data to subset data, create variables, validate and clean data.
3. **Analyze:** Further, analysis happens on this data. We can perform simple analyses like frequency and averages and complex analyses including regression and forecasting. SAS is the gold standard for statistical analyses.
4. **Present:** Finally we can present our analysis in the form of list, summary and graphic reports. We can either print these reports, write them to data file or publish them online.

What is SAS? What are the functions does it performs?

SAS means Statistical Analysis System, which is an integrated set of software products.

- Information retrieval and data management
- Writing reports and graphics
- Statistical analytics, econometrics and data mining
- Business planning, forecasting, and decision support
- Operation research and Project management
- Quality Improvement
- Data Warehousing
- Application Development

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List down the reasons for choosing SAS over other data analytics tools.

Answer: We will compare SAS with the popular alternatives in the market based on the following aspects:

SAS over data analytics tools	
Ease of Learning	SAS is easy to learn and provides an easy option (PROC SQL) for people who already know SQL.
Data Handling Capabilities	SAS is on par with all leading tools including R & Python when it comes to handling huge amount of data and options for parallel computations.
Graphical Capabilities	SAS provides functional graphical capabilities and with a little bit of learning, it is possible to customize on these plots.
Advancements in Tool	SAS releases updates in a controlled environment, hence they are well tested. R & Python, on the other hand, have an open contribution and there are chances of errors in the latest developments.
Job Scenario	Globally, SAS is the market leader in available corporate jobs. In India, SAS controls about 70% of the data analytics market share

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What are the advantages of using SAS?

There are many alternatives for SAS but what makes it unique as compared to others is:

Ease to understand: The concepts included in SAS are extremely easy to learn. Besides, it offers the most convenient option for those who already are aware about the SQL. On the other hand, R comes with a steep learning cover which is considered to be a low level programming language.

Data Handling Capacities: it is at par the most leading tool which also includes the R& Python. When it comes to handle the huge data, it is the best platform to choose.

Graphical Capacities: it comes with functional graphical capacities and has a limited learning scope. It is possible to customize the plots.

Better tool management: It helps in a release the updates with regards to the controlled environment. This is the main reason why it is well tested. Whereas if you considered R &Python, it has open contribution and risk of errors in the current development are also high.

What are the uses of SAS?

SAS software provides tools for a wide variety of applications in business, government, and academia. Major uses of SAS are economic analysis, forecasting, economic and financial modeling, time series analysis, financial reporting, and manipulation of time series data.

How do dates work in SAS?

The central to every data set is the data itself. In SAS, the data is available in form of tabular manner where there are some of the variables that occupy the column space and also the row space gets occupied with the observation section.

The numbers are treated by **SAS** as the numeric data while other things come under the character data. This is the main reason why SAS comes with two types. If you take an example, SAS date is the numerical value which is equivalent to the days from the 1st of January, 1960

What is the SAS data set?

A SAS data set is a file consisting of two parts.

- A descriptor portion
- A data portion

What is the basic structure of a SAS program?

SAS programs consists of :

- DATA step, which recovers & manipulates data.
- PROC step, which interpret the data.

What is PROC in SAS?

- In SAS, PROC steps analyze and process data in the form of an SAS data set. It controls a library of routines that perform tasks on SAS data set options such as sorting, summarizing and listing.

What is the basic syntax of a SAS program?

The points important for running SAS program are :

1. DATA statement, which names your data set.
2. The names of the variables in your data set are described by INPUT statement.
3. Statement should be ended through semi-colon(;).
4. Space between word and statement should be there.

What are the features of base SAS system?
It provides Ipv6 support, new true type fonts, extended time notations, restart mode, universal printing, checkpoint mode and ISO 8601 support.

Data Type

How many data types are there in SAS?

Answer: There are two data types in SAS. Character and Numeric. Apart from this, dates are also considered as characters although there are implicit functions to work upon dates.

If a variable contains letters or special characters, can it be numeric data type?

Answer: No, it must be character data type.

Informat and Format

State the difference between INFORMAT and FORMAT ?

- INFORMAT: To indicate SAS that a number should be read in a particular format
- FORMAT: To indicate SAS how to print the variable
- **Format**: i.e., WORDIATE18 and WEEKDATEW

Informat: i.e., comma, dollar, and date (MMDDYYw, DATEw, TIMEw, and PERCE

Difference between Informat and Format

Informats read the data while Formats write the data. Informat - To tell SAS that a number should be read in a particular format. For example: the informat mmddyy6. tells SAS to read the number121713as the date December 17, 2013. Format - To tell SAS how to print the variables.

What is SAS informats?

SAS INFORMATS are used to read, or input data from external files known as Flat Files ASCII files, text files or sequential files). The informat will tell SAS on how to read data into SAS variables.

Name types of category in which SAS Informats are placed

SAS informats are placed in three categories,

- Character Informats : $INFORMATw
- Numeric Informats : INFORMAT w.d
- Date/Time Informats: INFORMAT w.

ERRORs

What SAS features do you use to check errors and data validation?

To check errors, use the Log, and for data validation use things like Proc Freq, Proc Means or sometimes Proc print to see how data looks.

Why Is Sas Considered Self-documenting?

SAS is considered self documenting because during the compilation time it creates and stores all the information about the data set like the time and date of the data set creation later No. of the variables later labels all that kind of info inside the dataset and you can look at that info using proc contents procedure.

How do dates work in SAS data?

Data is central to every data set. In SAS, data is available in tabular form where variables occupy the column space and observations occupy the row space.

- SAS treats numbers as numeric data and everything else falls under character data. Hence SAS has two data types numeric and character.
- Apart from these, dates in SAS are represented in a special way compared to other languages.

Date	SAS Date Value
January 1, 1959	-365
January 1, 1960	0
January 1, 1961	366
January 1, 2003	15706

Dates in SAS

- A SAS date is a numeric value equal to the number of days since January 1, 1960.

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- Apart from Date Values, there are many tools to work on dates such as informats for reading dates, functions for manipulating dates and formats for printing dates.

What does PROC print, and PROC contents do?

To display the contents of the SAS dataset PROC print is used and also to assure that the data were read into SAS correctly. While PROC CONTENTS display information about a SAS dataset.

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